

Allozyme genetic diversity among *Fagus grandifolia* trees resistant or susceptible to beech bark disease in natural populations

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Abstract: American beech (*Fagus grandifolia* Ehrh.) trees resistant (R = 760) and susceptible (S = 681) to beech bark disease were located and mapped in nine natural stands in West Virginia (WV), Massachusetts (MA), Maine (ME), Nova Scotia (NS), and Prince Edward Island (PEI). Dormant bud tissue collected from each tree was examined by isozyme analysis to characterize the population genetic structure of R and S subpopulations mapped in each of four intensively sampled (R/S) stands, and five additional populations in which only R trees were sampled. Seventeen enzymes (with 9 polymorphic and 14 monomorphic loci) were analyzed to estimate variation across the spatial range of disease occurrence. All populations possessed significant levels of inter- and intra-population diversity; several parameters appeared to increase in magnitude from southwest to northeast across the range. In the R/S stands, percent polymorphic loci averaged 38%, average number of alleles/locus was 1.8 (3.0 for polymorphic loci), and mean expected heterozygosity was 0.165. Most genetic variation resided within populations (97%; $G_{ST} = 0.03$). Analyses of pooled R and S subpopulations demonstrated that observed heterozygosities were up to 26% higher in S trees, i.e., R tree populations exhibited consistent heterozygote deficits.

Résumé : Les auteurs ont localisé et cartographié des hêtres américains (*Fagus grandifolia* Ehrh.) résistants (R = 760) et susceptibles (S = 681) à la maladie corticale du hêtre au sein de neuf peuplements naturels de la Virginie de l'Ouest (WV), du Massachusetts (MA), du Maine (ME), de la Nouvelle-Écosse (NS) et de l'Île-du-Prince-Édouard (PEI). L'analyse isoenzymatique a été réalisée à partir des tissus de bourgeons en dormance récoltés sur chaque arbre afin de caractériser la structure génétique de population des sous-populations R et S cartographiées pour chacun de quatre peuplements échantillonnés intensivement (R/S), ainsi que de cinq populations additionnelles pour lesquelles des arbres de type R seulement ont été échantillonnés. Les auteurs ont analysé 17 enzymes (codant pour neuf loci polymorphes et 14 loci monomorphes) afin d'estimer la variabilité au sein de l'aire d'occurrence de la maladie. Toutes les populations ont démontré des niveaux significatifs de diversité inter- et intra-population; la magnitude de plusieurs paramètres semblait augmenter en progressant du sud-ouest vers le nord-est de l'aire de répartition naturelle. Pour les peuplements R/S, le pourcentage moyen de loci polymorphes était de 38%, le nombre moyen d'allèles par locus était de 1,8 (3,0 par locus polymorphe) et l'hétérozygotie espérée moyenne affichait une valeur de 0,165. La plus grande partie de la diversité génétique résidait au sein des populations (97%; $G_{ST} = 0,03$). Les analyses combinant l'ensemble des sous-populations R et S ont permis de démontrer que les niveaux d'hétérozygotie observée étaient jusqu'à 26% plus élevés chez les arbres S. En d'autres termes, les populations d'arbres R montraient constamment un déficit en hétérozygotes.

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Introduction

Attacks on forest trees by insects and diseases and the responses to such attacks and to other environmental stresses are often nonrandom in nature. Frequently, groups of trees become infested, infected, defective, and slow in growth or die. Group dying is usually attributed to localized occurrences of stress factors, and (or) of pests, or to the existence of localized or peculiar site factors that exacerbate stress–host–pest relationships. While many studies have examined

these relationships, relatively little attention has focused on how inherent genetic variability of host trees and spatial distribution of particular genomes may affect occurrence of insects and diseases.

Fagus grandifolia Ehrh. is one of a relatively few tree species that reproduce both by seed and vegetatively by root sprouts (Ward 1961). Although seeding must be the mechanism for initial establishment, root sprouting appears to be the main mode of regeneration in certain areas of its range (Abrell and Jackson 1977; Forcier 1975). Root sprouting in hardwood trees assures the continued development of stems in species where seed production may be low, erratic, subject to heavy predation or where embryos may be nonviable. Root sprouting also perpetuates specific genomes, increasing and spreading them through the forest community. This has advantages when the traits are desirable, such as disease or insect resistance, but is disadvantageous when the perpetuated trait is disease susceptibility.

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Fig. 1. Resistant (above, right) and susceptible (above, left) American beech trees in a natural stand located in Maine.

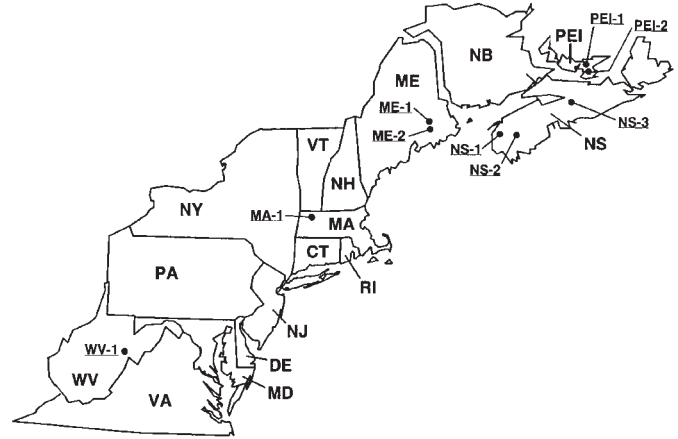


Beech bark disease causes significant mortality and defect in *F. grandifolia*. The disease occurs when bark, fed on by the beech scale, *Cryptococcus fagisuga* Lind., is subsequently attacked and killed by the bark fungi *Nectria coccinea* var. *faginata* Lohman, Watson & Ayers, and *Nectria galligena* Bres. (Ehrlich 1934; Lohman and Watson 1943; Cotter 1977; Houston 1994a). The conditions created by the disease, directly through parent tree mortality, or indirectly through salvage operations appear to have encouraged root sprouting of affected trees, and therefore, an increase in numbers of susceptible stems (Houston 1975).

Following its accidental introduction into Nova Scotia from Europe around 1890, *C. fagisuga*, and the disease it initiates, has continually spread westward and southward through the Maritimes (by the 1930s), Maine (by the 1940s), Massachusetts (by the 1950s), and into New York and Pennsylvania where its range is still expanding. Isolated outbreaks were discovered in West Virginia in 1980, in Ohio in 1984, and in Ontario in 1981. In 1993, the disease was described in North Carolina and Tennessee, some 400 mi south of the West Virginia outbreak (1 mi = 1.6 km) (Houston et al. 1979; Bisessar et al. 1985; Mielke et al. 1985; Houston 1994b).

Some trees remain insect- and disease-free in stands affected by beech bark disease (Fig. 1), and challenge trials have shown them to be resistant to *C. fagisuga* (Houston 1982, 1983). Resistant trees have been located in forests of Prince Edward Island (PEI), Nova Scotia (NS), Maine (ME),

Fig. 2. Locations of nine American beech stands sampled for isozyme analysis: WV-1, Monongahela National Forest, Randolph County, W.Va.; MA-1, Monroe State Forest, Berkshire County, Mass.; ME-1, Champion stand no. 2, T34 MD, Hancock County, Me.; ME-2, Champion stand no. 9, Osborn Twp., Hancock County, Me.; NS-1, Sissiboo Falls, Digby County, N.S.; NS-2, Kempt, Annapolis County, N.S.; NS-3, Riversdale, Colchester County, N.S.; PEI-1, St. Patrick Road (no. 1), Kings County, P.E.I.; PEI-2, Klondike, Kings County, P.E.I.



New Hampshire (NH), Massachusetts (MA), New York (NY), and West Virginia (WV). In all of these areas, resistant trees are found in relatively low numbers (ca. <1% of all beech trees in stands examined), and occur both as single individuals and, frequently, in discrete groups.

We hypothesize that (H_1) geographic sources of American beech do not differ in their genetic structure as characterized by a variety of population genetic statistics; (H_2) resistant and susceptible populations of American beech identified in our field surveys do not differ in their genetic structure; and (H_3) resistant populations of American beech in a variety of northeastern forests impacted by beech bark disease for different periods of time do not differ in their genetic structure.

The determination of acceptance or rejection of these hypotheses is based on the preponderance of evidence, from both available, commonly applied statistical tests, and inferential data.

Materials and methods

We examined the genetic variation patterns in *F. grandifolia* occurring in natural, mixed-species, northern hardwood stands located in WV, MA, ME, NS, and PEI (Fig. 2). Resistance to beech bark disease was used as a marker of unconfirmed genetically related individuals and groups (cohorts) of beech within stands impacted by the disease. Resistance of some, but not all, of these "clean" trees had been verified previously in challenge trials (Houston 1982). This determination was followed by sampling buds of all resistant (R) trees, and for four stands (WV-1, MA-1, ME-2, and NS-1), all susceptible (S) trees located in a 0.2- to 0.4-ha subplot containing a number of the resistant trees.

Isozyme analysis was used to characterize genetic patterns and parameters within and among stands for (i) beech in general, and (ii) the resistant and susceptible components of each population. Isozyme variation patterns of the resistant trees located in five other stands in ME, NS, and PEI were also examined. Enzyme systems (both polymorphic and monomorphic) used to study the

genetic structure of these beech populations were those that were available to us using current foliar enzyme extraction techniques, and not necessarily ones that may be implicated in disease or insect resistance in plants.

Four stands (populations) intensively sampled for both resistant and susceptible trees (i.e., R/S stands)

Stands (Fig. 2) were defined to include at least one group (subplot) of resistant trees (free of beech scale and associated cankering by *Nectria* spp.) and to encompass a minimum of 0.2 ha. For statistical analyses, resistant trees located within the stand, but outside the intensively sampled subplot, were also included. Analyses were also conducted that included only those resistant trees located within the boundaries of the intensively sampled subplots that contained the susceptible trees (trees with beech scale populations or *Nectria* cankering). The locations of all beech trees, ≥ 10 cm DBH (≥ 5 cm DBH in WV), in each stand were mapped.

Five stands (populations) sampled for resistant trees only

For the other five populations (ME-1, NS-2, NS-3, PEI-1, PEI-2; Fig. 2), only resistant trees were identified, mapped, and sampled as described above. Stands ranged in size from less than 0.10 ha to approximately 10.0 ha.

Sample collection, handling, and electrophoresis

Dormant buds were collected from the upper crown of each tree, stored under ice, frozen in liquid N₂ within 72–120 h, and held at -70°C until bud tissue extracts could be prepared for electrophoresis. Enzymes were extracted from bud tissue by removing bud scales and grinding single buds in 0.75 mL grinding buffer (Cheliak and Pitel 1984).

Using starch gel electrophoresis (Houston and Houston 1994), a total of 17 enzymes, including 9 polymorphic loci of 7 enzyme systems (phosphoglucose isomerase (*Pgi-2*)², phosphoglucomutase (*Pgm-1*)³, 6-phosphogluconate dehydrogenase (*6Pg-1*)², alcohol dehydrogenase (*Adh-2*)², malate dehydrogenase (*Mdh-1*)³, cytochrome oxidase (*Cto-1*)², peroxidase (*Per-3*)⁵) from dormant bud tissue were analyzed. An additional 14 loci (13 enzyme systems) including phosphoglucose isomerase (*Pgi-1*)², alcohol dehydrogenase (*Adh-1*)², triosephosphate isomerase (*Tpi*)⁶, isocitrate dehydrogenase (*Icd*)², α -galactosidase (α -*Gal*)², menadione reductase (*Mnr*)³, superoxide dismutase (*Sod*)⁴, glyceraldehyde-3-phosphate dehydrogenase (*G3p*)³, adenylate kinase (*Ak*)², fumerase (*Fum*)², glycerate-2-dehydrogenase (*G2d*)², peroxidase (*Per-1*)²,⁵ and aldolase (*Ald*)² demonstrated no polymorphism and were judged to be monomorphic.

All systems used 12.4% gels except for *Per* (12% gel). Following electrophoresis, enzymes were stained in 1.5-mm gel slices using protocols reported by Cheliak and Pitel (1984; *Cto*, *Mdh*, *Mnr*, *Pgm*), Siciliano and Shaw (1976; *Adh*, *Ak*, *Ald*, *Fum*, α -*Gal*, *Icd*, *6pg*, *G2d*, *G3p*, *Pgi*), Yen and Sadanaga (1977; *Per*), Soltis and Soltis (1989; *Tpi*), and Baum and Scandalios (1979; *Sod*). Isozyme banding patterns for isozyme loci were characterized by measuring r_f values for each isozyme band of each sample. Patterns for each enzyme system were verified by repetitive electrophoresis and by co-electrophoresis of individual tree samples. A total of 1441 trees (760 resistant; 681 susceptible) were analyzed.

For scoring purposes, the most anodal loci and alleles were designated 1, with the more cathodal alleles designated 2, 3, etc. Nu-

merous additional bands were observed for several enzyme systems, but difficulties in staining intensity, repeatability, and interpretation precluded their use.

It was not possible to confirm patterns of inheritance for these enzymes because of the lack of control-bred or half-sib progenies. Genetic control of polymorphic enzymes was inferred, however, from observation of functional subunit numbers in heteromultimer types of putative heterozygotes, and from published reports of work with these enzymes in other angiosperms.

Each tree was then assigned an isozyme "genotype" for each enzyme locus, and these genotypes were pooled to form a multilocus (i.e., composite) numerical enzyme genotype for each tree. Assignment of these alphabetic and numerical genotypes to each mapped tree permitted a visual assessment of the stand genetic architectures, including putative clonal structures (e.g., Houston and Houston 1994).

Statistical analyses

Analyses of population genetic structure were accomplished using the BIOSYS-1 program of Swofford and Selander (1989) as modified by W.C. Black (personal communication 1992).

Genetic diversity levels were estimated for individual stands (populations) as well as across all stands. The following common gene diversity measures were calculated: (i) percent polymorphic loci, (ii) average number of alleles per locus, and (iii) expected and observed heterozygosity. Percent polymorphic loci was calculated by dividing the number of polymorphic loci by the total number of loci scored. A locus was considered polymorphic if more than one allele was detected. Average number of alleles per locus was obtained by dividing the total of the alleles over all loci by the number of loci. Finally, the expected heterozygosity at Hardy-Weinberg equilibrium, H_{exp} (Nei 1978) and the observed heterozygosity, H_{obs} , were calculated for each locus and averaged over all loci.

Nei's (1973, 1977) parameters of gene diversity (H_T , H_S , D_{ST}) were used to investigate the distribution of genetic diversity within and among stands and in the population as a whole. Total gene diversity (H_T , the expected heterozygosity when all populations are treated as a panmictic unit), calculated according to Nei and Chesser's (1983) formula for small population size, was partitioned into within- (H_S) and among-stand (D_{ST}) components, and the overall coefficient of differentiation (G_{ST}) was calculated as D_{ST}/H_T . The mean G_{ST} value over all stands was obtained by averaging across all loci.

Chi-square analyses were used to test the null hypothesis that observed allele frequencies in the sample populations were a random sample drawn from a single population (Workman and Niswander 1970). To examine deviations in heterozygotes or homozygotes from those expected under Hardy-Weinberg equilibrium, Wright's F statistics (fixation indices: F_{IT} , F_{IS} , F_{ST} ; Wright 1943, 1951, 1965, 1969) were calculated according to the methods of Weir and Cockerham (1984). These fixation indices describe the effect of population structure on heterozygote frequencies.

Nei's (1972) genetic identity (and genetic distance) was calculated to measure allelic differentiation among stands and was the basis for clustering of stands by the unweighted pair-group method (UPGMA) using arithmetic means (Sneath and Sokal 1973).

All loci were included in the analyses of genetic variation, except where indicated. In general, recent recommendations by Berg and Hamrick (1997) regarding inclusion of monomorphic versus polymorphic loci in specific analyses were followed.

²Siciliano and Shaw (1976).

³Cheliak and Pitel (1984).

⁴Ridgeway et al. (1970).

⁵Kristjansson (1963).

⁶Ashton and Braden (1961).

Populations (Fig. 2) were grouped for analysis and hypothesis testing as follows:

- (1) H_1 : All trees (resistant and susceptible, pooled by stand) in four intensively sampled (R/S) stands; populations WV-1, MA-1, ME-2, NS-1;
- (2) H_{2a} : All resistant versus all susceptible trees (pooled across stands) in the four R/S stands; populations WV-1, MA-1, ME-2, NS-1;
- H_{2b} : Resistant trees versus susceptible trees within each of the four R/S stands; populations WV-1, MA-1, ME-2, NS-1;
- (3) H_3 : Resistant trees only in nine stands; populations WV-1, MA-1, ME-1, ME-2, NS-1, NS-2, NS-3, PEI-1, PEI-2.

The rationale for these groupings was to develop estimates of variation (i) across the spatial range of American beech now being impacted by beech bark disease, (ii) among populations of American beech judged to be resistant or susceptible to beech bark disease, and (iii) within stands where the disease has been present for varying lengths of time. Earlier analyses, which compared two populations (WV-1 and MA-1) affected more recently, explored the effects of clonality on genetic population structure (Houston and Houston 1994).

Results

Seven polymorphic enzyme systems yielded nine loci and 30 putative alleles. "Null" alleles (i.e., those that were not expressed (showed no activity) in gels for an enzyme system in a given individual) were detected for *Cto-1* and *-2*, and *Per-3*. Fourteen enzymes and (or) loci (*Pgi-1*, *Adh-1*, *Per-1*, *Per-2*, *Icd*, *Sod*, *Tpi*, *Mnr*, *G2d*, *Fum*, *Ak*, *Ald*, α -*Gal*, and *G3p*) were found to be monomorphic across all populations. Thus, 38% of all loci tested were polymorphic.

H_1 : Geographic sources of American beech do not differ in their genetic structure as characterized by a variety of population genetic statistics

Analysis of genetic variability in four stands sampled for both resistant and susceptible trees (R/S stands; populations WV-1, MA-1, ME-2, NS-1) revealed high levels of both intra- and inter-stand diversity (Table 1). Contingency χ^2 analyses demonstrated significant differences among stands for eight of nine loci. Within-stand genotype frequency distributions deviated significantly from Hardy-Weinberg equilibrium for *6Pg-2*, *Mdh-1*, *Cto-1*, *Cto-2*, and *Per-3* in all stands (Table 1). In each case, significant deviations were generally caused by excesses of heterozygotes at the *6Pg-2* and *Mdh-1* loci, and deficits of heterozygotes at the *Cto-2* and *Per-3* loci. There was, however, some variation from stand to stand in the loci involved. For example, *Cto-1* demonstrated an excess of heterozygotes in the WV-1 and MA-1 stands, but a deficit in the ME-2 stand. The *Pgm-1* locus had an excess of heterozygotes in the NS-1 stand, but was in equilibrium elsewhere.

The mean number of alleles per locus (A) varied from 1.70 in WV-1 to 1.91 in NS-1 (Table 2). The mean number of alleles, averaged over polymorphic loci only (AP), ranged from 2.78 in WV-1 to 3.33 in NS-1. Both measures tended to increase from southwest to northeast, as did the effective number of alleles (A_e). Percent polymorphic loci (P) ranged from 34.8% (MA-1) to 39.1% (WV-1, ME-2, and NS-1). Expected heterozygosity (H_e) also tended to increase as one moved northeast from WV-1 (0.150) to ME-2 (0.179), but decreased slightly in the NS-1 stand (0.172). Observed

heterozygosities (H_o) were also higher in the ME-2 and NS-1 stands.

Total genic diversity, H_T (Nei 1973, 1977), averaged over all populations and loci, was 0.434 (Table 3). For any given locus, the largest part of this diversity was attributable to H_S , the within-population diversity, with a mean value of 0.421. The proportion of diversity that resides within populations is thus H_S/H_T , and for these four stands it averaged 97%. Gene diversity between populations, D_{ST} , averaged 0.012. The coefficient of gene diversity, G_{ST} ($= D_{ST}/H_T$), a measure of the between-stand diversity level, averaged 0.025.

Wright's (1978) fixation indices indicated that these *F. grandifolia* stands were not in Hardy-Weinberg equilibrium; F_{IS} values varied widely across loci (Table 4). The mean (jackknife estimate across all loci) value of F_{IS} was 0.0110 (SE = 0.007), with values at or near zero for five of the nine polymorphic loci. The mean was strongly influenced by large negative values for *6Pg-2* and *Mdh-1*, and large positive values for *Cto-2* and *Per-3*, the latter due primarily to the influence of the null alleles observed at those loci. The same four loci demonstrated large deviations from expectations, suggesting the existence of disassortative mating within local populations. *Pgi-2* was the only polymorphic locus that did not differ significantly between stands. F_{IS} values were negative for six of nine polymorphic loci, but positive overall (avg. = 0.011). The mean F_{ST} value (0.0299; SE = 0.0006; jackknife estimate) confirmed that approximately 3% of the observed variability lies between stands.

The regression of genetic distance (D ; Nei 1978) on geographic distance was not significant (mean D = 0.077; r = 0.28, p = 0.588). Regression analyses of observed heterozygosity on latitude and longitude were also not significant (r = 0.47, p = 0.511). However, significance at the 0.10 level was noted for the regressions of expected heterozygosity on latitude (r = 0.93, p = 0.071) and on longitude (r = 0.92, p = 0.081).

A cluster analysis (Fig. 3) examined the relationship of the four R/S stands based on their genetic identities (Nei 1972). The ME and NS stands clustered together most closely, the WV stand differentiated next, and the MA stand was clearly the most highly differentiated from the others.

H_2 : Resistant and susceptible populations of American beech identified in our field surveys do not differ in their genetic structure

Analyses of resistant versus susceptible populations pooled from the four intensively sampled R/S stands (populations WV-1, MA-1, ME-2, NS-1) are summarized in Table 2. Values for A , AP , and P mostly mirrored those observed in the R/S stands. Observed heterozygosity was approximately 12% higher in the susceptible population, while mean expected heterozygosities were essentially equal. The effective number of alleles was equivalent in the pooled R and S populations, but was 50% higher than that for the R/S populations.

Wright's (1978) F statistics were very similar to those for the four R/S stands considered separately, as were Nei's (1973) gene diversity measures, except for G_{ST} (0.008), which was approximately one-third as large as that for the R/S stands. The lower estimate in this case is a result of a

Table 1. Allele frequencies^a, and χ^2 analyses of within- and among-population heterogeneity for *F. grandifolia* populations in West Virginia (WV-1), Massachusetts (MA-1), Maine (ME-2), and Nova Scotia (NS-1).

Locus	Allele	Population				Among stands ^b	
		WV-1	MA-1	ME-2	NS-1	χ^2	<i>p</i>
<i>Pgi-2</i>	A	0.079	0.090	0.121	0.116	5.36	0.147
	B	0.921	0.910	0.879	0.884		
	H-W equilibrium ^c						
	χ^2	1.12	0.325	3.05	2.524		
	<i>p</i>	0.291	0.569	0.081	0.112		
<i>Pgm-1</i>	A	0.145	0.358	0.302	0.211	85.61	0.000
	B	0.855	0.590	0.685	0.740		
	C	—	0.052	0.012	0.049		
	H-W equilibrium ^c						
	χ^2	3.40	4.239	1.04	10.774		
<i>6pg-1</i>	A	0.082	0.075	0.201	0.139	38.51	0.000
	B	0.918	0.925	0.799	0.861		
	H-W equilibrium ^c						
	χ^2	1.22	1.142	0.018	2.199		
	<i>p</i>	0.269	0.285	0.894	0.138		
<i>6pg-2</i>	A	0.313	0.358	0.326	0.328	18.36	0.005
	B	0.632	0.639	0.632	0.617		
	C	0.056	0.003	0.043	0.055		
	H-W equilibrium ^c						
	χ^2	51.72	54.83	87.63	112.04		
<i>Adh-2</i>	A	0.993	1.000	0.977	0.936	45.80	0.000
	B	—	—	0.018	0.041		
	C	0.007	—	0.005	0.023		
	H-W equilibrium ^c						
	χ^2	0.007	—	0.158	1.835		
<i>Mdh-1</i>	A	—	—	0.009	0.012	20.08	0.017
	B	0.493	0.431	0.470	0.473		
	C	0.319	0.347	0.336	0.369		
	D	0.188	0.223	0.185	0.144		
	H-W equilibrium ^c						
<i>Cto-1</i>	A	0.372	0.350	0.292	0.324	40.31	0.000
	B	0.372	0.494	0.384	0.450		
	O ^d	0.257	0.156	0.324	0.226		
	H-W equilibrium ^c						
	χ^2	293.43	259.62	351.76	583.08		
<i>Cto-2</i>	A	0.135	0.251	0.381	0.306	324.44	0.000
	B	0.546	0.130	0.391	0.468		
	C	0.043	—	0.014	0.003		
	D	—	0.009	—	0.001		
	E	—	0.009	—	0.001		
<i>Cto-2</i>	O ^d	0.276	0.601	0.214	0.221	324.44	0.000
	H-W equilibrium ^c						
	χ^2	163.79	520.39	332.42	1171.59		
	<i>p</i>	0.000	0.000	0.000	0.000		

Table 1 (concluded).

Locus	Allele	Population				Among stands ^b	
		WV-1	MA-1	ME-2	NS-1	χ^2	<i>p</i>
<i>Per-3</i>	A	0.641	0.633	0.418	0.562	150.71	0.000
	B	0.003	0.142	0.085	0.105		
	C	0.293	0.121	0.407	0.302		
	O ^d	0.059	0.104	0.089	0.031		
H-W equilibrium ^c							
	χ^2	200.96	237.73	385.66	589.16		
	<i>p</i>	0.000	0.000	0.000	0.000		
Total						729.20	0.000

^aAllele frequencies for the 14 monomorphic loci are not included.

^bContingency χ^2 analysis of among-population heterogeneity.

^cChi-square test of within-population Hardy-Weinberg equilibrium.

^d"Null" allele.

Table 2. Genetic variability measures in *F. grandifolia* populations (R/S stands) in West Virginia (WV-1), Massachusetts (MA-1), Maine (ME-2), and Nova Scotia (NS-1), and in pooled R and S populations within the R/S stands.

	R/S populations					Pooled populations	
	WV-1	MA-1	ME-2	NS-1	Mean	Resistant	Susceptible
No. of trees	152	173	281	389	249	314	681
Mean no. alleles/locus (A) ^a	1.70 (0.21)	1.74 (0.25)	1.83 (0.24)	1.91 (0.29)	1.80	1.91 (0.29)	1.83 (0.24)
Mean no. alleles/locus (AP) ^a	2.78 (0.28)	2.89 (0.39)	3.11 (0.26)	3.33 (0.41)	3.03	3.33 (0.41)	3.11 (0.26)
Effective no. alleles ^b	1.18	1.19	1.22	1.21	1.20	1.77	1.78
% loci polymorphic ^c	39.1	34.8	39.1	39.1	38.0	39.1	39.1
Mean $H_{exp}/locus^d$	0.150 (0.050)	0.159 (0.052)	0.179 (0.055)	0.172 (0.053)	0.165	0.170 (0.055)	0.171 (0.053)
Mean $H_{obs}/locus^e$	0.153 (0.059)	0.150 (0.056)	0.175 (0.057)	0.174 (0.057)	0.163	0.154 (0.053)	0.172 (0.059)

Note: Standard errors are shown below values in parentheses.

^a(A) averaged over all loci, (AP) averaged over nine polymorphic loci.

^bEffective no. alleles = $1 - (1/1 - H_{exp})$; averaged over all loci.

^cA locus was considered polymorphic if more than one allele was detected; averaged over all loci.

^dUnbiased estimate of heterozygosity (Nei 1978); averaged over all loci.

^eDirect count estimate of heterozygosity; averaged over all loci.

lower D_{ST} value, a measure of the between subpopulation diversity, for the pooled populations ($D_{ST} = 0.001$). The consequence is a lower estimate for the within-population proportion of variation.

Analyses of resistant versus susceptible populations within each of the four intensively sampled R/S stands (populations WV-1, MA-1, ME-2, NS-1; Table 5) revealed most of the +12% differential in observed heterozygosity noted above in the pooled susceptible tree population was accounted for by the Sissiboo, NS (NS-1), and the Monroe State Forest (MA-1) populations. In the NS-1 stand, $H_O = 0.191$ in the susceptible subpopulation, compared with $H_O = 0.152$ for the resistant subpopulation, a +26% difference. In all four stands, the susceptible subpopulations consistently displayed higher H_O values.

F statistics also differed among these R and S subpopulations. A mean F_{IS} value of -0.036 (jackknife estimate over loci; SE = 0.035) in the four susceptible subpopulations,

compared with a mean F_{IS} value of 0.064 (jackknife estimate over loci; SE = 0.035) for the resistant subpopulations, indicated a within-population heterozygote excess in the susceptible components, and a deficit of heterozygotes in the resistant ones.

H₃: Resistant populations of American beech in a variety of northeastern forests impacted by beech bark disease for different periods of time do not differ in their genetic structure

Analysis of resistant trees only in these nine stands revealed no clear geographic trends in allele frequencies. Measures of genetic variability (Table 6) were similar to those observed in the analysis of the four R/S stands, although trends are not as clear cut.

Total genic diversity of resistant trees (H_T) averaged over all populations and loci was 0.432, almost identical to that observed for the four R/S stands (0.434), although variability

Table 3. Nei's (1973) statistics of gene diversity for (i) nine polymorphic loci in R/S populations (WV-1, MA-1, ME-2, NS-1) and (ii) pooled across polymorphic loci of trees in nine resistant populations of *F. grandifolia*.

Locus	H_T	H_S	D_{ST}	G_{ST}
<i>Pgi-2</i>	0.182	0.181	0.001	0.003
<i>Pgm-1</i>	0.420	0.403	0.016	0.039
<i>6Pg-1</i>	0.218	0.212	0.005	0.024
<i>6Pg-2</i>	0.492	0.491	0.001	0.002
<i>Adh-2</i>	0.046	0.045	0.001	0.021
<i>Mdh-1</i>	0.630	0.628	0.002	0.003
<i>Cto-1</i>	0.650	0.642	0.007	0.011
<i>Cto-2</i>	0.673	0.615	0.058	0.087
<i>Per-3</i>	0.591	0.569	0.022	0.037
Mean	0.434	0.421	0.012	0.025
SE	0.077	0.074	0.006	0.009
Pooled across polymorphic loci in nine resistant populations				
	0.432	0.408	0.024	0.056

Table 4. Fixation indices (Wright 1978; Weir and Cockerham 1984) for nine polymorphic loci in four populations (WV-1, MA-1, ME-2, NS-1) of *F. grandifolia*.

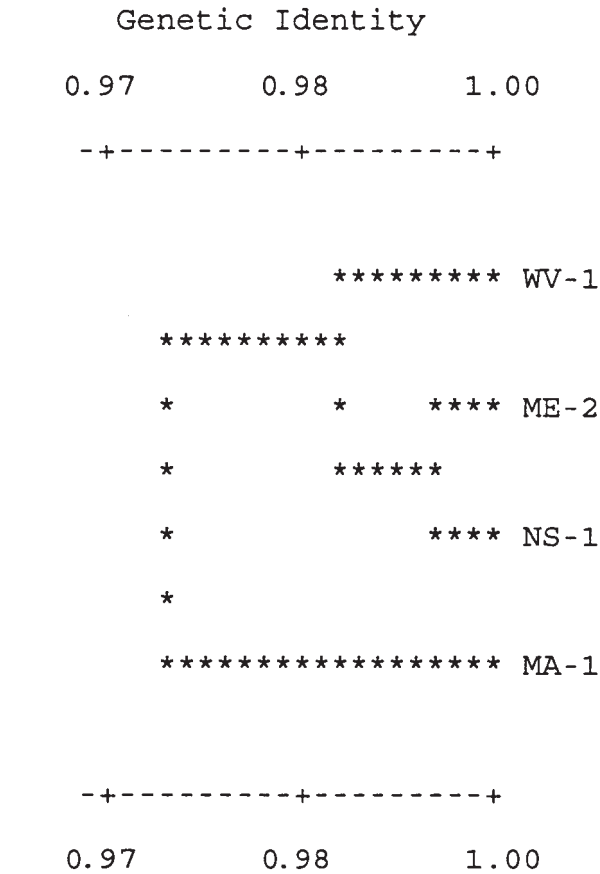
Locus	Fixation index		
	Total (F_{IT})	Within stand (F_{IS})	Among stand (F_{ST})
<i>Pgi-2</i>	-0.0662	-0.0682	0.0018
<i>Pgm-1</i>	0.0239	-0.0133	0.0367
<i>6pg-1</i>	-0.0227	-0.0489	0.0249
<i>6pg-2</i>	-0.3877	-0.3884	0.0005
<i>Adh-2</i>	-0.0189	-0.0419	0.0221
<i>Mdh-1</i>	-0.5057	-0.5094	0.0025
<i>Cto-1</i>	0.0421	0.0317	0.0107
<i>Cto-2</i>	0.4192	0.3628	0.0885
<i>Per-3</i>	0.5848	0.5674	0.0402
Mean	0.0379	0.0088	0.0294
Jackknife estimates over all loci			
Mean	0.0423	0.0110	0.0299
SE	0.0074	0.0071	0.0006

appeared to be apportioned somewhat differently. The within-population proportion of diversity (H_S/H_T) was lower (94% vs. 97%), and G_{ST} , the between-stand diversity, averaged 5.6%, twice that obtained in the R/S stand analysis (Table 3).

Contingency χ^2 analyses demonstrated highly significant differences among stands for all nine polymorphic loci ($\chi^2 = 1283.8$, $p = 0.001$), indicating that these stands differed significantly in their allele frequencies. Within-stand genotype frequency distributions also deviated significantly from Hardy-Weinberg equilibrium for a majority (64%) of loci, although this varied from stand to stand. For example, *Pgi-2* deviated from Hardy-Weinberg equilibrium only in NS-3 and PEI-1, and *Pgm-1* only in WV-1, while *Per-3* was in conformance only in NS-3.

F statistics (Wright 1978), calculated according to Weir and Cockerham (1984), resulted in a mean $F_{IT} = 0.059$, indi-

Fig. 3. Cluster analysis diagram, based on genetic identity (Nei 1972), of four intensively sampled *Fagus grandifolia* stands. Samples include resistant and susceptible trees.



cating a slightly higher level of heterozygote deficit (6%) across all stands, considered as a whole, as compared with the four R/S stands (4%). The mean F_{IS} value (0.024) suggested the within-stand heterozygote deficit was higher in these populations than in the R/S stands (0.011), while the mean F_{ST} value (0.036) was about equal (0.030).

Discussion

H_1 : Variation among R/S stands

Average levels of diversity noted in the *F. grandifolia* populations included in this study, as characterized by numbers of alleles per locus, percentage of polymorphic loci, and estimates of heterozygosity, are lower than those reported in the literature for most other hardwood species, including *Fagus sylvatica* L. (Comps et al. 1990; Müller-Stark 1985, 1989a, 1989b), Japanese beech, *F. crenata* Blume (Takahashi et al. 1994), and sand-shinnery oak, *Q. havardii* Rydb. (Mayes et al. 1998). There were instances where particular alleles were present in some stands, but not in others. In most cases, these represented alleles that were present at very low frequencies, and thus (i) may have escaped detection in some stands by chance, or (ii) may represent private alleles, peculiar to the specific stands in which they were found. Average numbers of alleles per locus were also lower than those reported for other long-lived woody perennials, e.g., *Quercus*, 2.8 (Hokanson et al. 1993); *Populus*, 2.4

Table 5. Genetic variability at 23 loci in beech bark disease-resistant and disease-susceptible components of *F. grandifolia* populations WV-1, MA-1, ME-1, NS-1.

Population			No. of alleles per locus (AP) ^a	% loci polymorphic ^b	Mean heterozygosity	
					Obs. ^c	Exp. ^d
WV-1	R ^e	32	2.22 (0.36)	26.1	0.148 (0.064)	0.126 (0.050)
	S	120	2.67 (0.24)	39.1	0.155 (0.058)	0.154 (0.050)
MA-1	R	36	2.89 (0.39)	34.8	0.126 (0.053)	0.155 (0.054)
	S	137	2.56 (0.29)	34.8	0.156 (0.057)	0.157 (0.051)
ME-2	R	79	3.00 (0.24)	39.1	0.174 (0.055)	0.191 (0.056)
	S	202	3.11 (0.26)	39.1	0.176 (0.059)	0.172 (0.055)
NS-1	R	167	3.22 (0.43)	39.1	0.152 (0.054)	0.160 (0.052)
	S	222	3.11 (0.26)	39.1	0.191 (0.063)	0.175 (0.051)

Note: Standard errors are shown below values in parentheses.

^aAveraged over nine polymorphic loci.

^bA locus was considered polymorphic if more than one allele was detected.

^cDirect count estimate of heterozygosity.

^dUnbiased estimate of heterozygosity (Nei 1978).

^eR, resistant trees; S, susceptible trees.

Table 6. Genetic variability in nine beech bark disease-resistant *F. grandifolia* populations in West Virginia, Massachusetts, Maine, Nova Scotia, and Prince Edward Island.

	Population									
	WV-1	MA-1	ME-1	ME-2	NS-1	NS-2	NS-3	PEI-1	PEI-2	Mean
No. of trees	32	36	124	79	167	44	66	180	32	—
No. alleles/locus (A) ^a	1.48 (0.19)	1.74 (0.25)	1.78 (0.23)	1.78 (0.23)	1.87 (0.28)	1.70 (0.19)	1.70 (0.22)	1.96 (0.30)	1.57 (0.19)	1.73
No. alleles/locus (AP) ^a	2.22 (0.36)	2.89 (0.39)	3.00 (0.24)	3.00 (0.24)	3.22 (0.43)	2.78 (0.15)	2.78 (0.32)	3.44 (0.44)	2.44 (0.29)	2.86
Effective no. of alleles ^b	1.14	1.18	1.21	1.24	1.19	1.20	1.18	1.23	1.18	1.19
% loci polymorphic ^c	26.1	34.8	39.1	39.1	39.1	39.1	34.8	39.1	30.4	35.7
Mean $H_{exp}/locus^d$	0.126 (0.050)	0.155 (0.054)	0.175 (0.055)	0.191 (0.056)	0.160 (0.052)	0.170 (0.052)	0.153 (0.051)	0.185 (0.056)	0.150 (0.050)	0.163
Mean $H_{obs}/locus^e$	0.148 (0.064)	0.126 (0.053)	0.167 (0.054)	0.174 (0.055)	0.152 (0.054)	0.180 (0.058)	0.163 (0.057)	0.180 (0.058)	0.183 (0.064)	0.164

Note: Standard errors are shown below values in parentheses.

^a(A) averaged over all loci; (AP) averaged over nine polymorphic loci.

^bEffective number of alleles = $1 - (1/1 - H_e)$; averaged over all loci.

^cA locus was considered polymorphic if more than one allele was detected.

^dUnbiased estimate of heterozygosity (Nei 1978).

^eDirect count estimate of heterozygosity.

(Jelinski and Cheliak 1992); *Robinia*, 2.6 (Surles et al. 1989). The number of alleles per locus increased from southwest to northeast; a trend that may be tied to increasing population sizes, although observed heterozygosities did not follow this trend as closely. Müller-Starck et al. (1992) summarized allozyme studies of variation in *F. sylvatica*, finding

an average for observed heterozygosities of 27%. While it was not clear whether monomorphic loci were included in calculations for the *F. sylvatica* studies, observed heterozygosity in *F. crenata* Blume (Takahashi et al. 1994) averaged 16% using 14 loci, including two that were monomorphic. Observed heterozygosity in our four R/S stands averaged

16.3%, less than that found for European beech but identical to that noted in Japanese beech.

Genic diversity values were similar to those found for species with similar life histories. The observed mean G_{ST} value for the four R/S stands (0.025), indicating that approximately 3% of the total variability was accounted for by the between-stand component, is identical to that observed in six subpopulations of *Populus tremuloides* (Jelinski and Cheliak 1992), and lower than that noted for widely separated populations of *Gleditsia triacanthos* ($G_{ST} = 0.059$; Schnabel and Hamrick 1990), both clone-forming species. Overall gene diversities (G_{ST}), measured in several studies with *F. sylvatica* and summarized by Müller-Stark et al. (1992), averaged 0.038 (range = 0.016–0.054), indicating a 50% higher level of stand differentiation in that nonclonal species than the average for our populations of *F. grandifolia* (avg. $G_{ST} = 0.025$; SE = 0.009; range = 0.002–0.087). Because many stands of *F. sylvatica* in western Europe were established as plantations, it is possible that the stands referred to by Müller-Stark et al. (1992) originated from widespread provenances, resulting in relatively high diversity between stands occurring over a relatively small area. Somewhat higher values (avg. = 0.049) were reported for *Nothofagus truncata* (Colenso) Cockayne in New Zealand (Haase 1992), a species whose populations, because of their mode of origin and (or) physical separation, may have become genetically divergent owing to founder effects or isolation by distance (Haase 1993).

Computation of Wright's (1978) fixation index (mean $F_{ST} = 0.0299$; jackknife estimate) allowed comparison with several other studies employing this measure. Cuguen et al. (1988) and Comps et al. (1990), evaluating fewer loci across many populations of *F. sylvatica*, reported between-stand variabilities of 6.4% and 3.4% (mean), respectively. In a recent report by Merzeau et al. (1994), adult and juvenile trees were sampled in three stands of *F. sylvatica*. Multilocus F_{ST} values (mean = 0.027) were very similar to those we report here.

Some pairs of stands that were separated by larger geographic distances had smaller genetic distances (e.g., WV and NS; $D = 0.048$) than other pairs separated by smaller geographic distances (e.g., WV and MA; $D = 0.104$). In addition, allelic frequencies did not appear to follow any regular pattern related to latitude or stand location. In *F. sylvatica*, however, peroxidase allelic frequencies were found to be correlated with altitude and moisture regime (Comps et al. 1990, 1991).

Cluster analysis (Fig. 3) of the four R/S stands suggested a pattern of relationships among these populations that could be explained partly on the basis of geographic relationships, and partly on ecological and environmental factors. While the reasons for the marked differentiation of the MA stand are not entirely clear, factors related to its origin and history of disease encounter may be involved. This stand was the least severely impacted by the disease, and its genetic and age structure (Houston and Houston 1994) imply a relatively recent origin from abandoned farmland, in contrast with the other stands in this study which are known to have been in continuous forest cover. Genotype spatial patterns indicate that the majority of the trees in this stand had developed from individual seeds (Houston and Houston 1994). The rel-

atively low number of clonal groups in the MA stand compared with other stands suggests that root-sprout triggering disturbances have been less frequent than in the long-affected stands.

In summary, given the evidence of differences among the geographic sources of American beech included in this study as indicated by significant variability among stands in allele frequencies; consistent geographic trends in mean number of alleles/locus (A and AP), and expected and observed heterozygosities; significant departures from Hardy-Weinberg equilibrium; the significant regression of expected heterozygosity on latitude and longitude; and the separation of stands based on cluster analysis, we reject the null hypothesis H_1 , and conclude that these stands differ in their genetic structure.

H₂: Variation among resistant and susceptible (sub)populations

Resistant and susceptible tree "populations" (pooled from the four R/S stands) were very similar for most allele frequencies and genetic parameters, except that observed heterozygosity was higher in the susceptible population.

F statistics computed for the four R/S stand resistant and susceptible tree subpopulations also suggested differences between these groups. An F_{IS} value of -0.036 for the four susceptible subpopulations versus $+0.064$ for the four resistant subpopulations indicates an excess of heterozygotes in the susceptible components, and a deficit of heterozygotes in resistant trees. Also, the coefficient of gene differentiation, G_{ST} , was more than twice as high in the resistant population (0.068) than in the susceptible one (0.031).

Results of analyses of data derived from these two population partitions (i.e., (i) R and S tree subpopulations "pooled" from the four R/S populations, and (ii) R and S tree subpopulations "within" each R/S population) provide evidence for a determination of acceptance or rejection of the null hypotheses H_{2a} and H_{2b} . Specifically, an increase in AP values for R subpopulations from southwest to northeast; a twofold difference in G_{ST} values between pooled R and S populations; higher mean observed heterozygosity per locus in the pooled S subpopulation; and F_{IS} values that corroborate an excess of heterozygotes in S stand components and a heterozygote deficit in R subpopulations, lead us to reject the null hypotheses (H_{2a} and H_{2b}), and conclude the R and S subpopulations in the sampled stands differ in their genetic structure.

H₃: Variation among nine resistant populations

Genetic variability measures (Table 6) were mostly similar to those for the R/S stands. However, greater genic diversity ($G_{ST} = 0.056$; Nei 1973) among the nine resistant stands than among the four R/S stands might be expected, given the R/S stands were a more restricted geographic sample of the species range. The G_{ST} value for these populations was still lower, however, than the average reported by Hamrick and Godt (1990) for long-lived woody perennials ($G_{ST} = 0.076$).

Larger within-stand heterozygote deficits in these resistant groups of trees may reflect an underlying genetic difference between resistant and susceptible populations, related to adaptation, tree vigor, and stand history. Mean annual growth rates of trees susceptible to beech bark disease often

exceeded those of paired resistant trees prior to the onset of disease (Gove and Houston 1995). Similar growth patterns have been noted for disease-susceptible or disease-resistant populations in other species. Early growth by *Picea abies* (L.) Karst. trees heavily attacked by *Armillaria mellea* (Hrib et al. 1983), and Geils and Jacobi (1993) noted that *Pinus contorta* subsp. *latifolia* trees severely affected by *Cronartium comandrae* Peck were among the larger and faster growing trees in their respective stands prior to onset of disease-caused growth losses. Progeny of *Pinus monticola* trees selected for resistance to *Cronartium ribicola* tended to grow somewhat more slowly (–10%) than those of nonresistant trees, possibly a consequence of inbreeding (Bingham et al. 1973). Whether the slower annual growth rates of beech bark disease resistant trees is a consequence of inbreeding is not certain, but the consistent trend for resistant tree populations to have heterozygote deficits suggests this possibility.

In stands long-exposed to beech bark disease, many susceptible trees gradually become weakened, slow in growth, and eventually die. In such stands, higher vigor-resistant trees are good seed producers and provide the potential for the development of family groups composed of individuals related by descent at the half- or full-sib level. In a recent study of nut caching by blue jays (Johnson et al. 1997), jays cached nuts from each load separately, usually several metres apart. They flew to the first location but usually hopped to subsequent caching locations. Consecutive caching sites sometimes were over 100 m apart. If blue jays exhibit similar caching behaviors when transporting beech seeds, and apparently they do (Johnson and Adkisson 1985), these patterns of distribution may also contribute to the development of spatial substructuring in beech stands by creating clusters of related individuals, and thus help to explain the patterns of genotypes observed within these populations. Given (i) the root-sprouting habit of *F. grandifolia*, which can lead to the development of rather extensive putative clones within stands (Houston and Houston 1994), and (ii) the localized deployment of beech seeds by blue jays and other seed predators, assortative matings among ramets of the same clone or related clones, and among half or full siblings, are highly probable in this species.

Small F_{IS} values for the three NS and two PEI populations indicate there is little inbreeding within these populations, and that most of the overall genetic deviation (F_{IT}) is a result of heterogeneity in allele frequencies among populations. The average F_{IT} was five to eight times greater than the average F_{IS} , indicating moderate differentiation among stands.

Gene diversity parameters for the PEI stands were very similar to those for the three NS stands (Table 6), indicating that these populations were structured similarly. Populations in this part of the range exhibited the lowest total diversity (H_T), but some of the highest G_{ST} values, suggesting that they were among the most highly differentiated of the stands included in this study. They also exhibited the least apparent inbreeding, as reflected by the small F_{IS} values. Overall, the PEI and NS stands were much more similar to each other than to the ME stands in terms of the magnitude of the various gene diversity measures. The Canadian stands (and likely those in Maine) probably have been harvested or salvage-logged more frequently than others. Such activities

tend to promote a higher degree of clonality as a result of the disturbance of root systems by logging operations.

Total gene diversity was higher in ME (especially) and in populations to the southwest than in NS and PEI. The mean positive F_{IS} and F_{IT} values indicate an overall deficit of heterozygotes, perhaps caused by assortative mating between near neighbors. An alternative, but less likely, mechanism is suggested by the consistency in heterozygosity deficits in resistant populations, which implies that, if segregation of these allozyme loci is not independent, a recessive mode of resistance (i.e., selection against heterozygosity) may be the causal reason, rather than inbreeding. However, given the life history characteristics of this species, we believe that inbreeding is the more likely mechanism.

In summary, these statistics lead us to reject the null hypothesis (H_3) of no differences among resistant populations which have been impacted by beech bark disease for varying lengths of time by virtue of their geographic location in the species range. Specifically, contingency χ^2 analyses indicate highly significant differences between stands in allele frequencies for all loci. In addition, significant departures from Hardy–Weinberg equilibrium within stands for a majority of loci were noted.

Still, we do not assume, given the life history characteristics of this species, especially its vegetative reproduction characteristics and extreme shade tolerance, that temporal differentiation of beech populations caused by disease pressure has had sufficient time to develop, even in the longest affected forests of the Maritimes and Maine. The data does indicate, however, that these populations, like the R/S stands, differ from each other in their genetic structure. Additional work is needed to further clarify the complex relationships between beech bark disease impact and stand genetic structure.

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